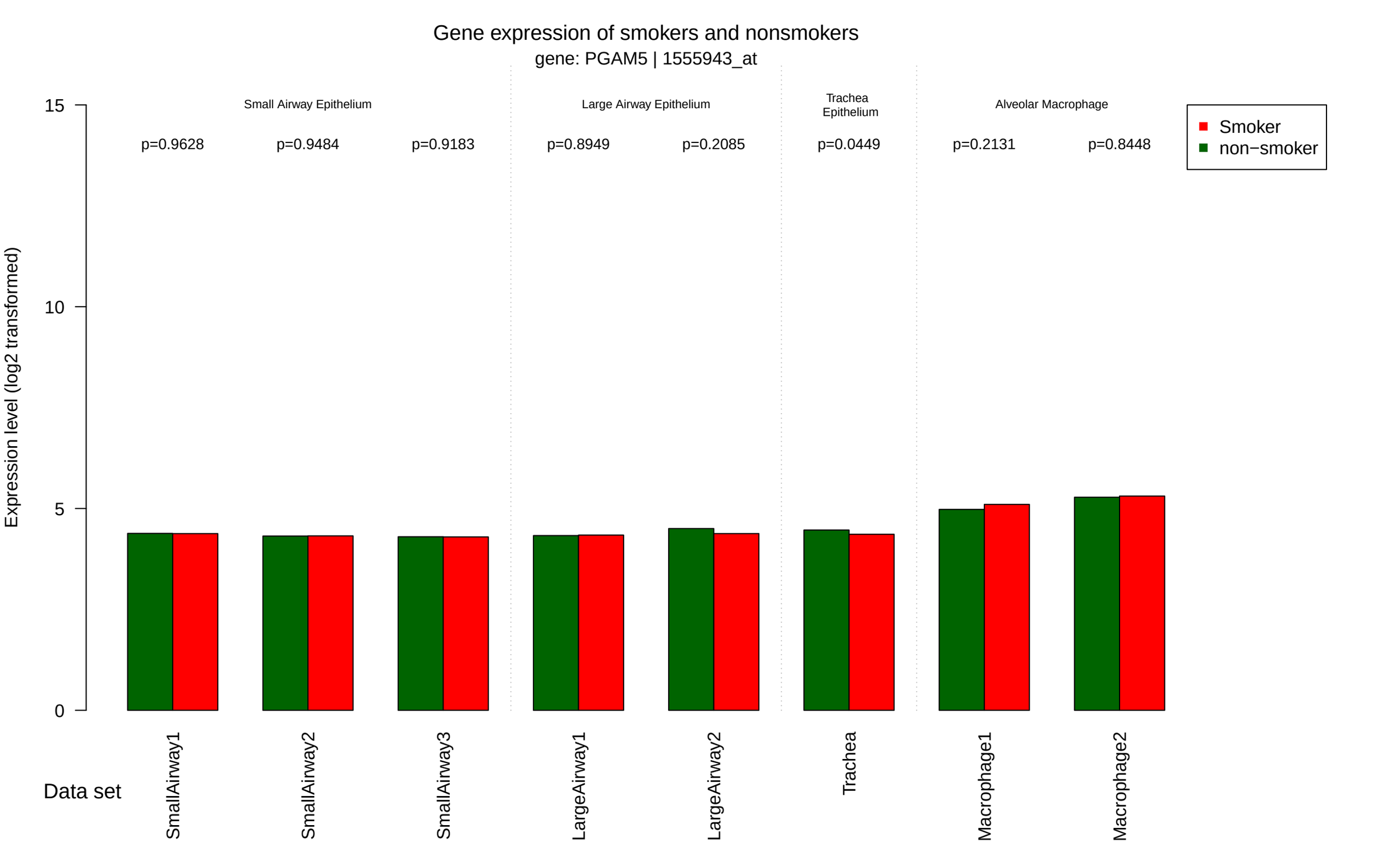
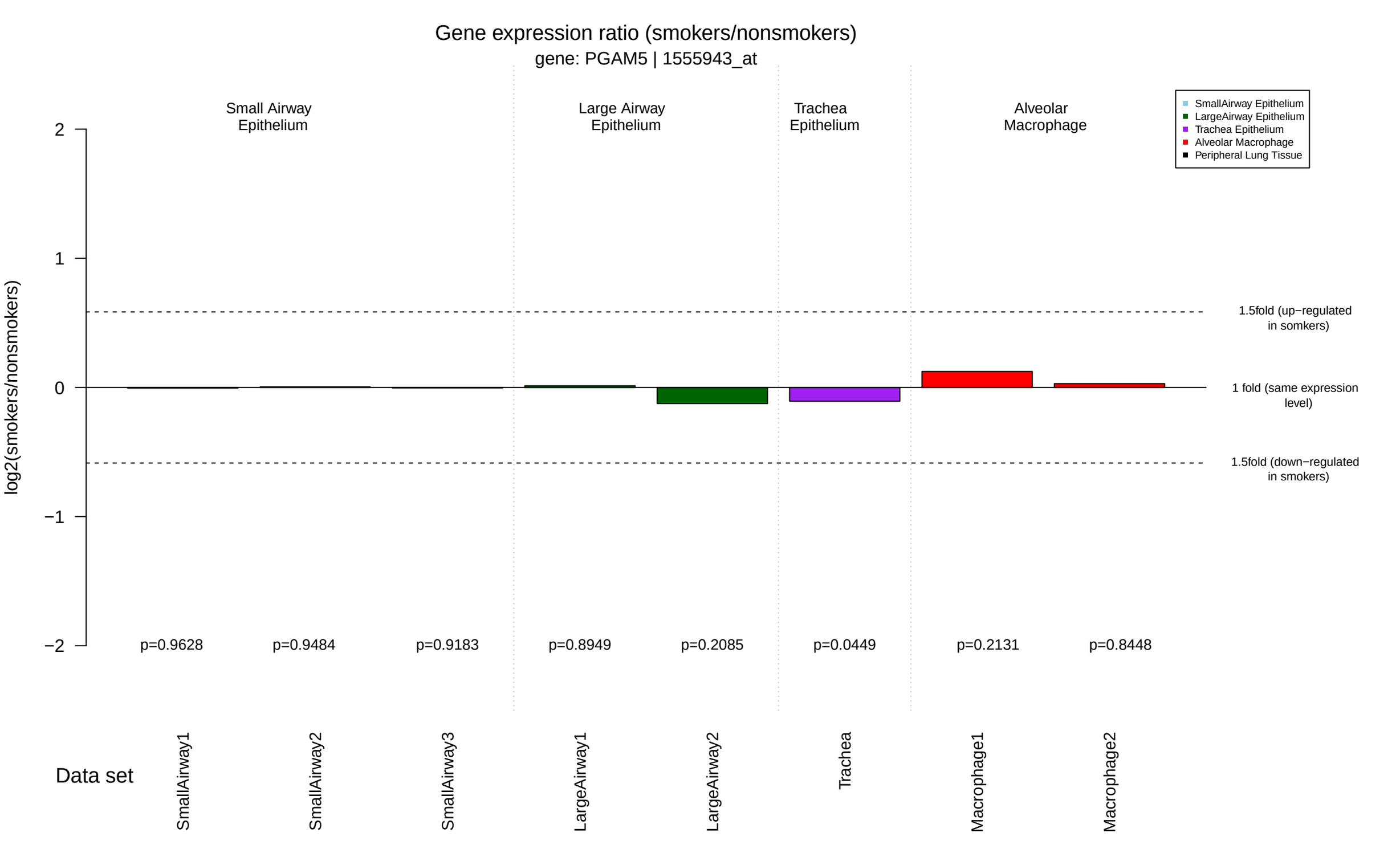
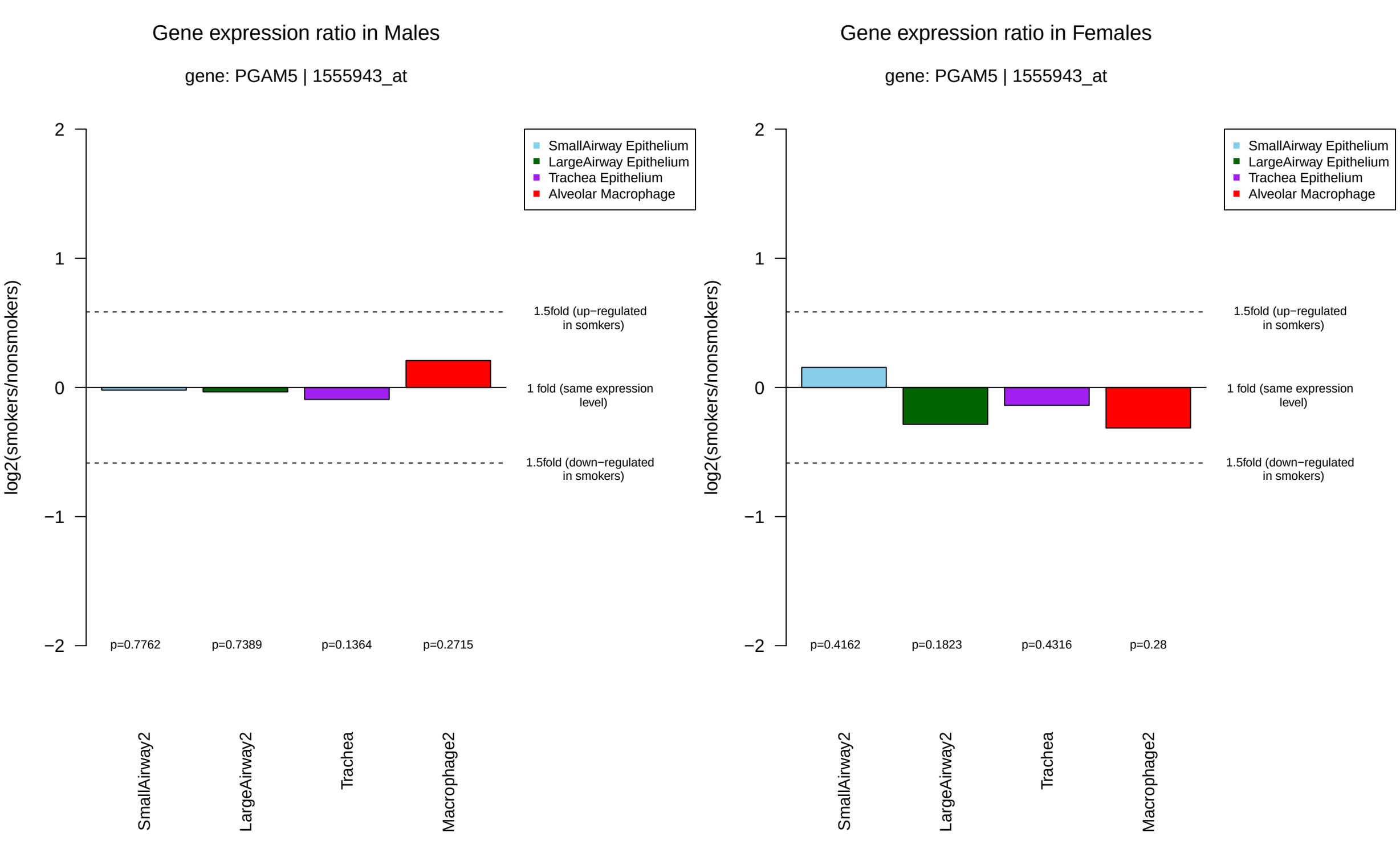
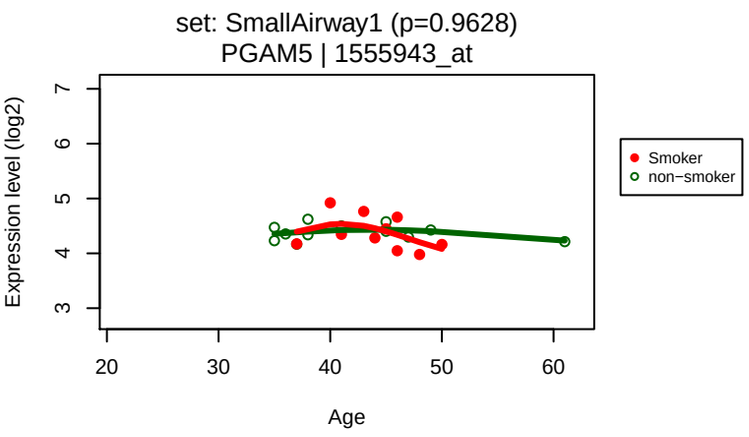
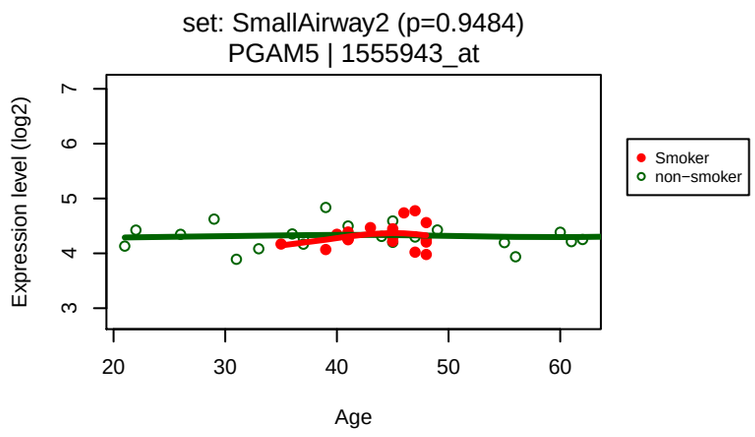
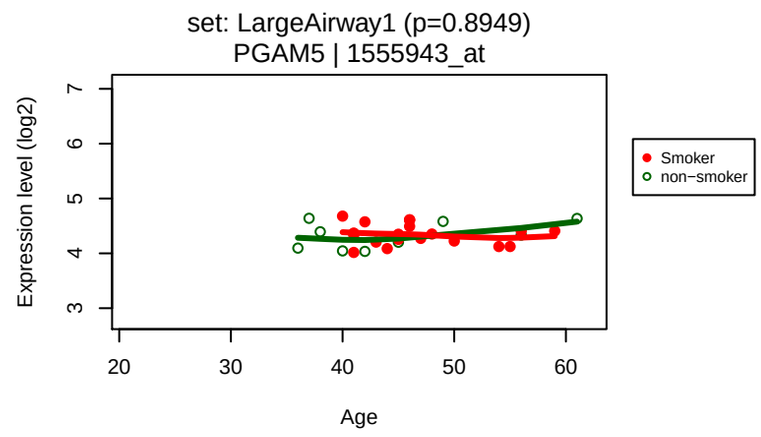

Gene expression of smokers and nonsmokers
gene: PGAM5 | 1555943_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9628
p=0.9484
p=0.9183
p=0.8949
p=0.2085
p=0.0449
p=0.2131
p=0.8448
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

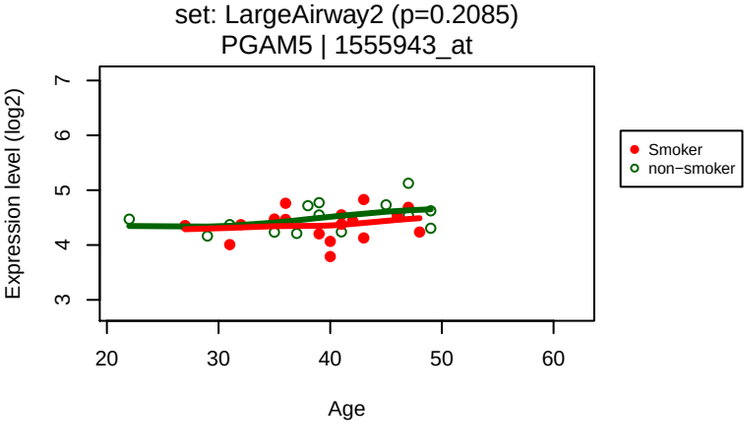

Gene expression ratio (smokers/nonsmokers)
gene: PGAM5 | 1555943_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9628
p=0.9484
p=0.9183
p=0.8949
p=0.2085
p=0.0449
p=0.2131
p=0.8448
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

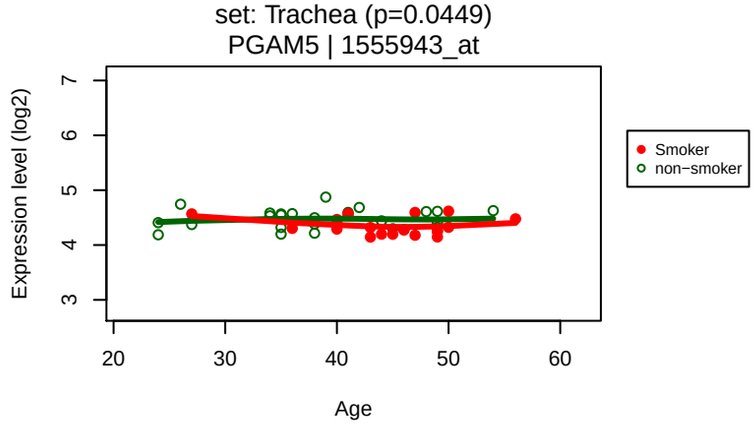

Gene expression ratio in Males
gene: PGAM5 | 1555943_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7762
p=0.7389
p=0.1364
p=0.2715
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PGAM5 | 1555943_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4162
p=0.1823
p=0.4316
p=0.28
SmallAirway2
LargeAirway2
Trachea
Macrophage2

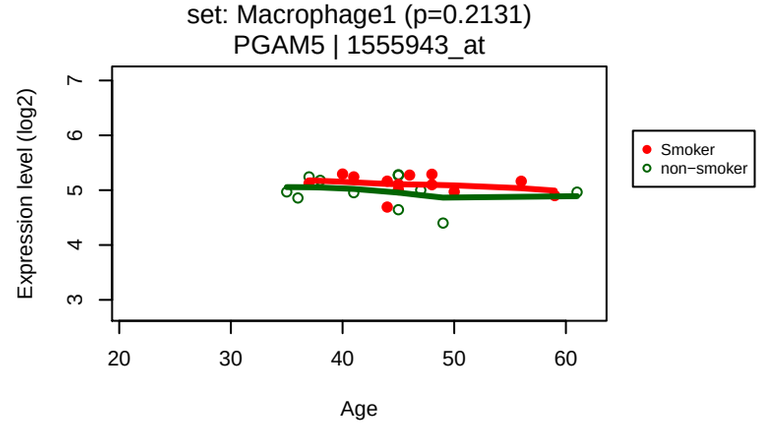

set: SmallAirway1 (p=0.9628)
PGAM5 | 1555943_at
Expression level (log2)
Age
Smoker
non−smoker

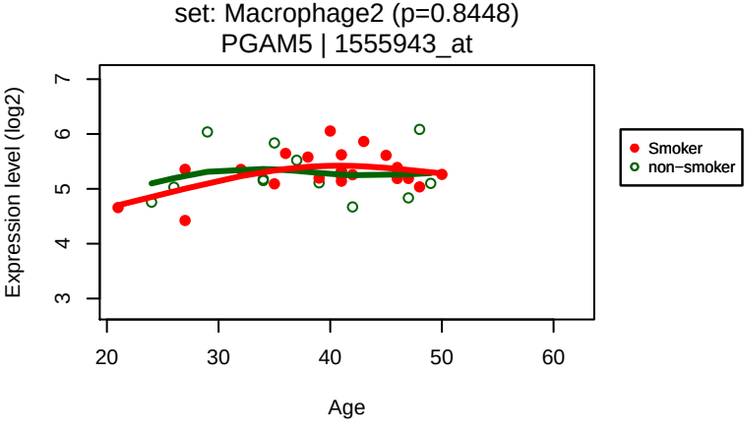

set: SmallAirway2 (p=0.9484)
PGAM5 | 1555943_at
Expression level (log2)
Age
Smoker
non−smoker

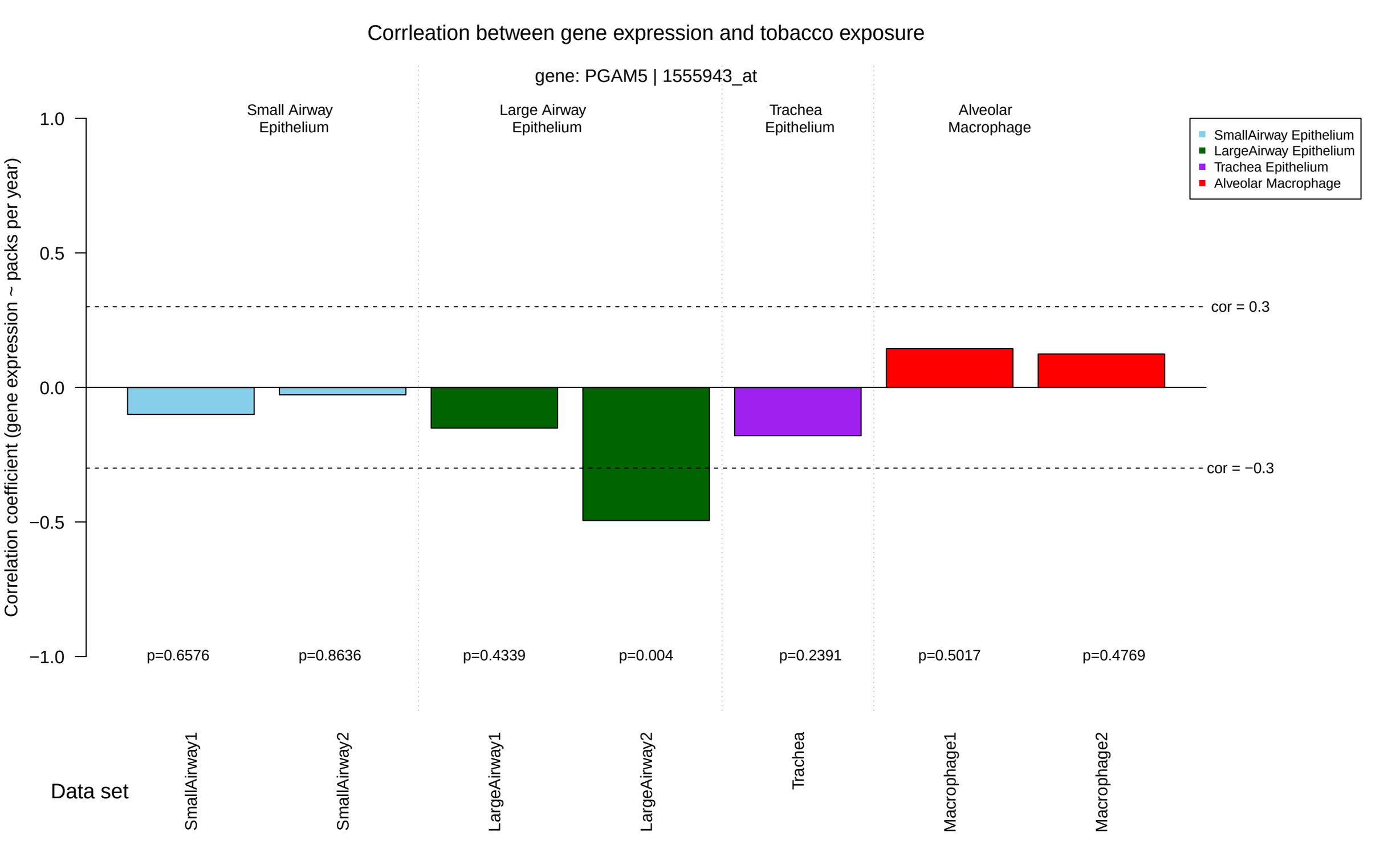

set: LargeAirway1 (p=0.8949)
PGAM5 | 1555943_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2085)
PGAM5 | 1555943_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0449)
PGAM5 | 1555943_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2131)
PGAM5 | 1555943_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8448)
PGAM5 | 1555943_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PGAM5 | 1555943_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6576
p=0.8636
p=0.4339
p=0.004
p=0.2391
p=0.5017
p=0.4769
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2